



EDM.320 – Key Design Decisions

CO-004-03 Workflow Design - MCPS

Montgomery County Public Schools

Oracle ERP Project – Phase 1



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1. Introduction

1.1 Overview

MCPS requires spending transactions to be approved by their respective owners. The owners are considered to be responsible for managing their budgeted funds. Operating & Capital Grant Funds are budgeted at various levels but can be broadly classified at an Organization (Dept) , Function (Program), Location, Category and Project levels. For every combination, there is an MCPS staff assigned to track the usage. These MCPS staffs are considered to be the owners of these funds and when spending transactions are created anywhere in the application, the respective owners must approve the transaction prior to the end of the life cycle of the transaction.

Oracle Cloud applications provide the ability to route spending transactions types (e.g. journal entries, requisitions, expense reports) through a pre-defined approval process referred to as workflow. A separate and unique workflow can be established for each transaction type.

The Systems Integrator has proposed the following:

S.No	Transaction Type	Organization Owner	Function + Location + Operating Project Owner	Other Approvals
1	Purchase Requisitions	HCM Department Cost Center Manager Approval	Role Based Approval	Based on Purchasing Policies
2	Purchase Orders & Agreements			Buyer Approvals as per Purchasing Policies
3	Expense Reports	HCM Department Cost Center Manager Approval	Function & Project Owner Approvals	Capital Projects are routed to Project Mangers.
4	AP Invoices	Resource Based Approvals	Resource Based Approvals	Based on Payables Policies
5	GL Journals	Role Based	Role Based	Based on Accounting

		Approval	Approval	Policies
6	Capital Grants	Project Role Based Approval		Based on Grants Administration Policies

The purpose of this document is to provide a functional overview of the workflow that will be established to support the Phase1 MCPS Financials Cloud implementation. Defining the logic for the routing of the various transaction types is a key component for the implementation.

The design contained herein is the result of the Workshop, Sprint1, and Sprint2 design sessions held between MCPS and AST module team members. Changing the design of these workflows could impact the project timeline.

1.2 Key Concepts

To understand the logic utilized for approval workflow, it is important to understand four key concepts:

- 1) **Cost Centers** – a cost center is a number that represents the smallest segment of an organization for which costs are collected and reported. It can be a division, location and/or project of an organization to which costs can be specifically allocated. Oracle requires at least one segment on the Chart of Accounts to be identified as “Cost Center”. MCPS Chart of Accounts has “Organization” Segment as the identified “Cost Center”.
- 2) **HCM Department** – a HR function that allows organizations to define a group of workers working towards common objectives. These departments can be assigned Managers who may control and maintain the department’s activities. Oracle allows association of one or more Cost Centers to these Departments and also allows each Cost Center to have its respective Manager. With MCPS HCM not currently being implemented in Oracle, the current implementation team will not create these HCM Departments. The implementation team will bring in the Capital Grants related Departments and assign their respective Cost Center Segment Value and associate their respective managers as well. The rest of the Departments would be grouped under a single “Central Office” related department where the rest of the Cost Center Segment Values along with
- 3) **HCM Cost Center Manager Approval** – based on the Cost Center Value charged in the spending transaction, the cost center manager is derived, and the transaction is routed for the Manager’s Approval

-
- 4) **Roles** – roles are job roles that represent system permissions that an employee has. The implementation team is using this concept to route the “Non-Cost Center” manager approvers. For eg. Anytime a Function Approver is required, a skeleton role for the function is created and assigned to the owner of the function segment. Based on the Function Value charged in the spending transaction, the MCPS person occupying the Function Approver role gets to approve such transactions.

2. MCPS Funds Owners Approvals

2.1 Background

The workflow change from EBS to Cloud inevitably creates implementation design challenges. For example, in FMS, the department segment became a floater segment which stored non-department values for non-standard approval routing. In Cloud ERP, the ability to add department or other values that were not predefined was eliminated. The requirements across various transaction entities is much more rigid.

In FMS, a combination of Approval Management and Workflow Builder tools were deployed. In Cloud ERP, Business Process Management (BPM) is the only dedicated tool for Workflow Approvals. This tool expects Organizations to implement a hierarchical approval for their transactions. The hierarchies are natively designed to build approvals based on “Who’s the transaction creator?”. MCPS requires that transactions to be routed based on “What’s the account on the transaction?”.

2.2 EBS Legacy Account Owner Workflow Logic

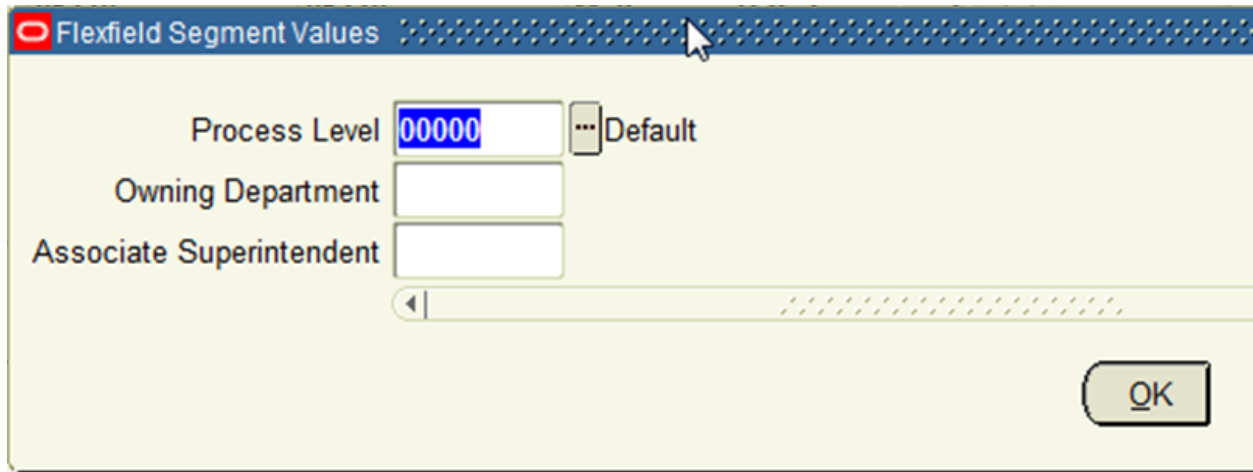
MCPS utilized descriptive flex fields (DFF) to assign approvers for a department. When a request for approval permissions was submitted, the Accounting Department would create new department permutations using the DFF and assign approval permissions to individuals. The creation of new department permutations, instead of selecting predefined departments from a list, resulted in an overabundance of department permutations and an unmanageable approval structure. This also diluted the purpose of the key segment value “Department” in the Chart of Accounts design.

MCPS has the following DFFs for reporting on Chart of Account (COA) accounts, which are utilized to facilitate reporting requirements and approval workflows.

- Process level – User defined level with a company, such as office, dept, or division. For most process levels within MCPS, the budgeting of positions is to the process level. Every process level has a number and a name associated with it.
- Owning Dept – Affects the following workflows for requisitions, expense reports, and journal entries (non-budget)
- Associate Superintendent – Impacts workflow for expense reports. Top level routing for

executives.

Department Segment



2.3 MCPS Requirements & Functionality Gap

MCPS approvals requires most of their approvals to be routed based on where the money is spent from. It is also determined that more than 90% of the transaction would require either a Department (Organization) approval or a Program (Function) Owner Approval. However, the solution should be elastic enough to be expanded for other levels of approvals as well.

Oracle Cloud ERP approval configuration engine is predominantly based on “Who spends the money” rather than “Where the money was spent from?”. The native behavior of the workflow engine is to build routing paths based on the person trying to spend and their HR Supervisory paths. With MCPS not maintaining the HR Supervisory information in Cloud, this option would not be applicable for MCPS.

The gap in the functionality is to have transactions routed for the owner of the fund as opposed to the native functionality of routing transactions based on Spender’s HR supervisory hierarchy.

To avoid some of the pitfalls identified above with the EBS account owner logic, MCPS has developed a workflow based on the following logic:

Seq. No	Chart of Accounts Segment	Rule	Action if True	Action if False
1	PROJECT	Approver Found	Stop Here	If not go to location

2	LOCATION	Approver Found	Stop Here	If not go to function
3	FUNCTION	Approver Found	Stop Here	If not go to organization
4	ORGANIZATION	Approver Found	Stop Here	

2.4 Cloud Based Account Owner Workflow Design

The combination of “Account Charged” + “Where the money is coming from?” requires a unique solution for MCPS transaction approvals. AST proposes the following solution:

1. Upon reviewing the workflow rules the following can be deduced:
 - a. 95% of the approvals are Organization Segment based
 - b. The remaining 5% requires either multi-segment or a different segment (function/project/location) approval
2. Create all Projects related Departments in HCM. Assign their respective Cost Center Values and Managers.
3. Create a skeleton application role for every department in the following format:

ORG_<Department Value>_APPROVER

4. Assign the respective approvers with the Organization Approver roles
5. For Functions requiring approvals, create skeleton application role in the following format:

FUNC_<Function Value>_APPROVER

6. Assign the respective approvers with the Function Approver Roles
7. For Operating Projects requiring approvals, create skeleton application role in the following format:

PROJ_<Project Value>_APPROVER

8. Assign the respective approvers with the Project Approver Roles
9. For Locations requiring approvals, create skeleton application role in the following format:

LOC_<Location Value>_APPROVER

10. Assign the respective approvers with the Location Approver Roles
11. These roles are skeleton with no function/duty assigned to them. These roles are defined just for routing transactions to the respective approvers.
12. In BPM for each transaction entity:

- a. Define multiple approval stages; one approval stage is created for each segment

approver.

- b. For Requisitions & Expenses, native cost center manager approval path will be implemented.
 - c. For AP Invoices & Journals, the cost center approval will also be a role based approval due to the application limitations. Transactions will be routed to “ORG_<Department Value>_Approver” Owners.
 - d. For transactions that require additional approval, have the BPM parse each segment value from the concatenated code combination value.
 - e. On each approval stage, route the transaction to the respective segment approver role.
 - f. Application routes transactions to the person occupying the respective segment approver role.
13. The reason for choosing this role-based approval design is predominantly because of the limitations and the inconsistent list of attributes across transactions available with the BPM tool.
14. The advantage of having a role-based approval design is relatively simpler maintenance during a re-org. After a re-org, revoke the role from the current approver and assign the role to the new approver.
15. Another advantage of this design is that if there are **additional exceptions** to be added. For example, to set up a new location requiring an approval one would:
- a. Open the BPM workflow,
 - b. Go to the location approval stage, and
 - c. Add the new location value to the condition.
16. The biggest disadvantage of this design is that upon inquiring the transaction for the approval tree, the results would display the roles that the transactions being routed to and not the person occupying the role. To bridge the information, AST proposes a custom report/screen to supplement the approval design. The design of this report/screen would be to allow users to key-in their transaction number and obtain the current approver’s name and their approval role.
17. An Approval Report is currently developed for the following Transaction Types:
- Requisition Approval

-
- GL Journals Approval
 - AP Expense Report Approval
 - AP Invoices Approval
 - PO Approval

Alternatively, All Transactions can be selected to display Approvers for all transactions.

17. Report screenshot:

approval Report

Transaction Type: AA

Approval Report

Apply

Automatic Zoom

Approval Report

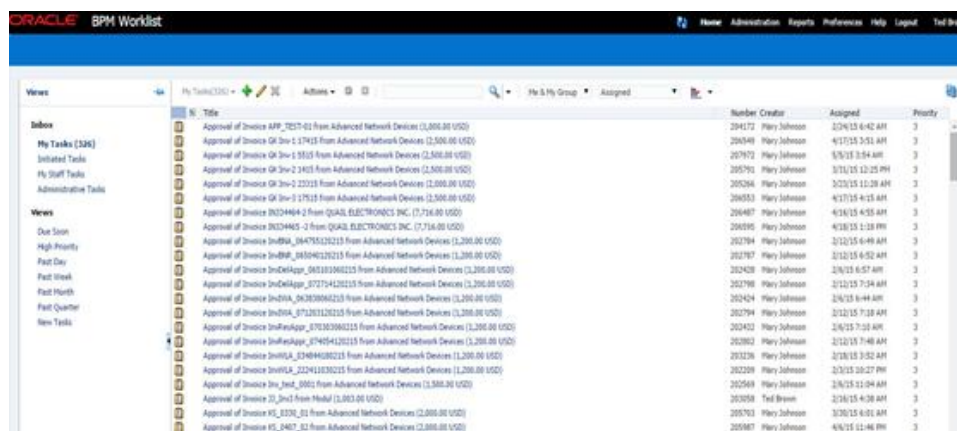
TRANSACTION TYPE	Trx Number	Current Approver	CREATOR	Trx Id
AP Expense Report	EXP0011990398	Mirek, Nicola	plandbar1@arkcorporation.com	300000011990398
AP Expense Report	EXP0012076821	Rita Eamba	bambant	300000012076821
AP Expense Report	EXP0012076979	Ranga Santhanam	bambant	300000012076979
AP Expense Report	EXP0012076993	Ranga Santhanam	bambant	300000012076993
AP Expense Report	EXP0012077003	Deborah Camp	bambant	300000012077003
AP Expense Report	EXP0012077003	Rita Eamba	susan.chen	300000012077003
AP Expense Report	EXP0012082107	Applications Development Framework Application Identity for CRM	mercort	300000012082107
AP Expense Report	EXP0012082574	Gomez, Dylan	luciland	300000012082574
AP Expense Report	EXP0012082588	Karen Rotons	luciland	300000012082588
AP Expense Report	EXP0012082594	Karen Rotons	luciland	300000012082594
AP Expense Report	EXP0012082608	Karen Rotons	luciland	300000012082608
AP Expense Report	EXP0012082635	Karen Rotons	noahroher	300000012082635
AP Expense Report	EXP0012198250	Terry Miller	noahroher	300000012198250
AP Expense Report	EXP0012198434	Chen, Susan	luciland	300000012198434
AP Expense Report	EXP0012198447	Rita Eamba	luciland	300000012198447

3. Cloud ERP Approvals Technical Design

3.1 BPM Overview

The BPM Worklist is a web-based application that enables users to view notifications and setup/manage approval rules.

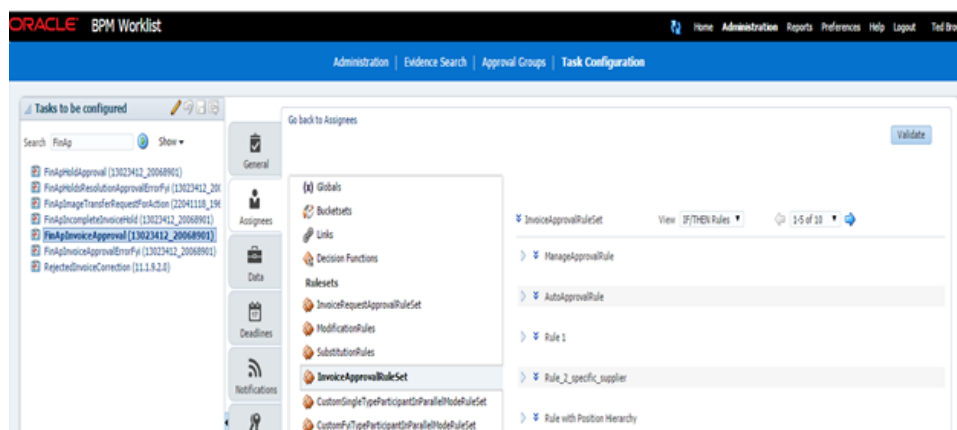
1. Users can view tasks assigned to them and perform various approval-related actions. For example, an approver can view the payables invoice approval notification, invoice details such as header amount, line details, amounts, and perform the necessary approval action.



The screenshot shows the Oracle BPM Worklist interface. The top navigation bar includes links for Home, Administration, Reports, Preferences, Help, and Logout. The main content area displays a list of tasks assigned to the user. The tasks are organized into a table with columns for Number, Creator, Assigned, and Priority. The tasks are listed in descending order of priority.

Number	Creator	Assigned	Priority
204172	Mary Johnson	2/24/15 6:42 AM	3
204349	Mary Johnson	4/17/15 3:51 AM	3
207972	Mary Johnson	5/6/15 3:54 AM	3
205791	Mary Johnson	3/11/15 12:25 PM	3
205364	Mary Johnson	3/23/15 11:28 AM	3
204053	Mary Johnson	4/27/15 4:15 AM	3
204487	Mary Johnson	4/14/15 4:55 AM	3
204095	Mary Johnson	4/18/15 1:38 PM	3
202784	Mary Johnson	2/12/15 6:49 AM	3
202787	Mary Johnson	2/12/15 6:52 AM	3
202428	Mary Johnson	3/6/15 6:57 AM	3
202798	Mary Johnson	2/12/15 7:34 AM	3
202424	Mary Johnson	3/6/15 6:44 AM	3
202794	Mary Johnson	2/12/15 7:18 AM	3
202423	Mary Johnson	3/6/15 7:18 AM	3
202862	Mary Johnson	2/12/15 7:46 AM	3
203236	Mary Johnson	2/18/15 3:52 AM	3
202209	Mary Johnson	2/12/15 10:27 PM	3
202569	Mary Johnson	3/6/15 11:04 AM	3
203058	Ted Brown	2/18/15 4:38 AM	3
205763	Mary Johnson	3/20/15 4:01 AM	3
205987	Mary Johnson	4/6/15 11:46 PM	3

2. For those with appropriate permissions, approval rules can be set up and managed.



The screenshot shows the Oracle BPM Worklist interface in the 'Task Configuration' section. The left sidebar contains a list of tasks to be configured, including 'InvoiceApproval' and 'InvoiceApprovalError'. The main content area displays the configuration for the 'InvoiceApprovalRuleSet'. The configuration includes a list of rulesets, such as 'ManageApprovalRule', 'AutoApprovalRule', 'Rule 1', 'Rule 2_specific_supplier', and 'Rule with Position Hierarchy'. The 'Rule 2_specific_supplier' rule is currently selected.

3.2 Approval Key Concepts

3.2.1 List Builders

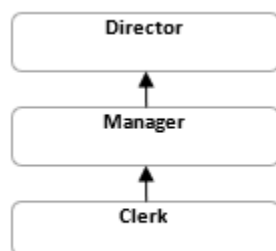
List builder is the way approvals management builds the list of approvals required for a transaction based on the rule condition. Each approval rule is associated with a list builder for generating the list of approvers. For Payables invoice approvals, the following list builders are supported:

- Supervisory
- Job Level
- Position
- Approval Group
- Resource

3.2.2 Supervisory

Approvals can be set up based on the employee supervisory hierarchy, which is defined in HCM. Employees must be set up in HCM with appropriate jobs and supervisors. For example, the clerk reports to the manager, who reports to the director.

The following figure is an example of an employee and supervisor approval structure.



For complete details on setting up the employee supervisory hierarchy, refer to the HCM application.

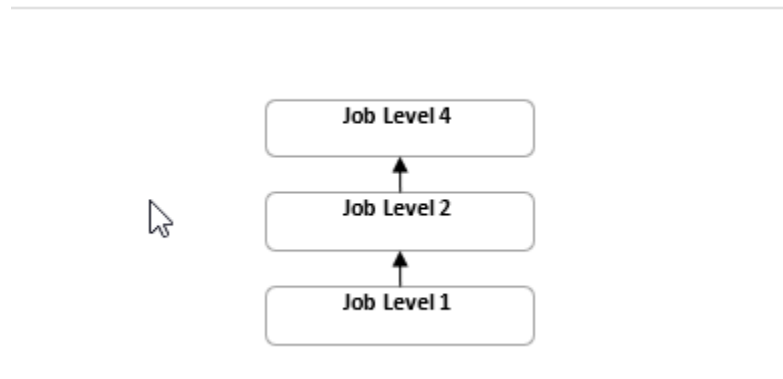
3.2.3 Job Level

Job level routings are based on the supervisory hierarchy defined in HCM. Employees must be set up in HCM with the appropriate job levels and supervisors. For example, Job Level1 employee, a clerk, reports to Job Level2 employee, a manager, who reports to Job Level4 employee, a director.

The approval list is generated based on the starting position specified in the rule and continues until an approver with a sufficient job level is found. The supervisory hierarchy needs to be defined along with the corresponding job levels.

NOTE: As part of the HCM implementation, MCPS must consider translating the current Approval Routing to a Job Codes & Job Levels and avoid the complexity that is currently implemented in this Phase.

The following figure provides an example of a Job Level structure:



For complete details on setting up job levels and assigning job levels to employees, refer to the HCM application.

3.2.4 Position

Organizations can also choose to route invoice approvals based on the position hierarchy defined in HCM. The position hierarchy needs to be defined and employees must be assigned the corresponding positions.

Here's an example of a position hierarchy approval structure for Beta Retail:



For complete details on setting up the position hierarchy, refer to the HCM application.

3.2.5 Approval Groups

An approval group consists of a static predefined set of users configured to act on a task. Depending on the participant type defined, approval tasks are routed to an approval group in serial or parallel mode. For example, you can create an approval group called Finance Group comprised of users from the finance department who need to participate in the approving of a task.

New approval groups can be created, or existing approval groups can be edited from the Approvals Groups tab on the BPM Worklist application Administration page.

Steps to create an approval group:

Sign into the BPM Worklist application as an administrator

1. Click the Administration link.

2. Click the Approval Groups tab.
3. Click the  icon in the left side pane.
4. Enter the name of the approval group.
5. Click the  icon in the Details area to add users to the approval group.
6. Select users to add to the group.
7. Save the approval group.



3.2.6 Resource

Using the Resource list builder, you can build the approvers list by using a specific user, group duty role, or application role. Resource list builder is one of the easiest ways to build the approvers list.

3.2.7 Participant

A participant is a user or set of users who are participating in the approval process, including their routing policy in the process. Each participant is associated with a single rule set. Approval rules are defined in the context of a rule set and you can create many rules under one rule set. Based on your approval requirements, you need to configure the approval rules within the context of a participant.

3.2.8 Types of participants

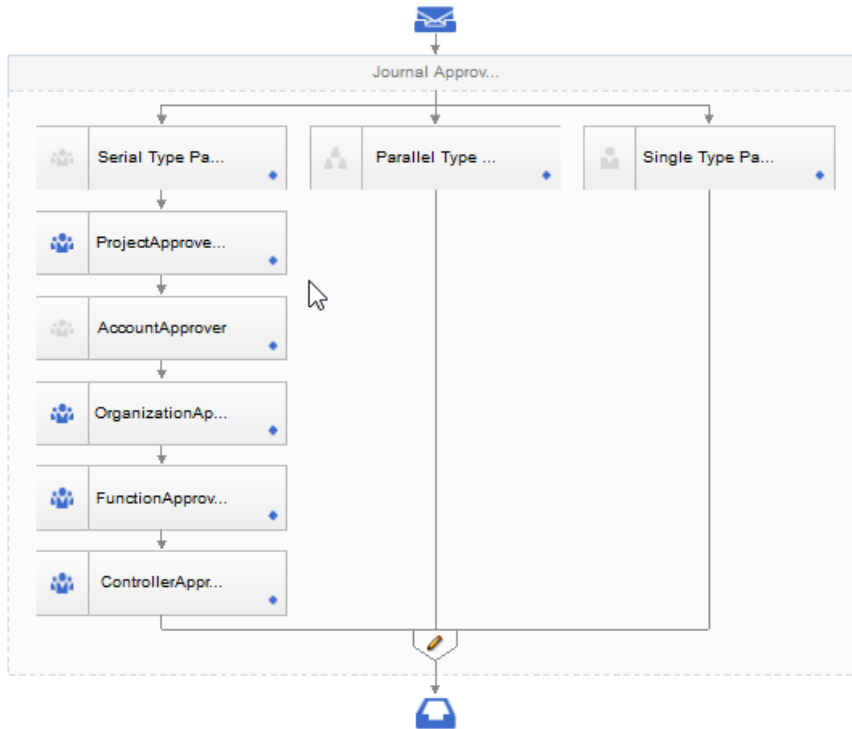
Participant	Description
Serial	Approvers work in sequence. This is a commonly used participant type for configuring rules using management hierarchies where approvals are performed in sequential fashion one after another.
Parallel	A set of approvers work in parallel. For example, an invoice is assigned to all users simultaneously. Unanimous approval is required; if one user rejects the invoice, the invoice will be rejected.
Single	A task is assigned to more than one user; however, the outcome is solely decided by the first user to approve or reject.
FYI	Users receive a notification for informational purposes only, no response is necessary. Users cannot perform any actions on FYI notifications.

3.3 Journal Workflow Example

Below is an example of BPM Workflow rule with screenshots. BPM Workflows may change based on business requirements. The screenshots are for basic understanding of BPM and may not be related to current business process.

High-level Diagram for BPM Approvals:

Assignees



3.3.1 Empty Approval Group

We have created an Empty Approval Group if the business conditions are not met. This will be required for Contra Approval rules.

The screenshot shows a software interface for configuring an approval group. On the left, under the 'Groups' tab, a search bar contains 'Approval Group' and a list below it shows 'Empty Approval Group' with a group icon. On the right, the 'Details' panel shows the group's name as 'Empty Approval Group' and a 'Members' section stating 'No Members Exist'.

3.3.2 Project Rule

The Project Rule specifies that if the segment7 (Project) in COA is "008815" , "008902" for Debit Line, Approval should go to user assigned to PROJ_<Segment7>_APPROVER Role.

ProjectApprover

IF

```

(for each case where) {
  Journal Line is a Journal Line and
  Journal Line Entered Debit isn't null and
  Journal Line Entered Credit is null and
  RL.string.substring(Journal Line.Account Combination,30,6) in "008815", "008902"
}

```

THEN

List Builder	Resource
Response Type	Required
Users	null
Participants	Groups null
Application Role	"PROJ_" + RL.string.substring(Journal Line.Account Combination,30,6) + "_APPROVER"
Rule Name	"ProjectApprover"
Dimension Id	null

We can see the User assigned to above role eg: PROJ_008815_APPROVER in below screen.

Edit Role PROJ_008815_APPROVER: Users

User Login	Display Name	Description
KALLJAJ		

1

2

3

4

5

6

7

3.3.3 Contra Project Rule

The Contra Rule specifies the action when the conditions are not met. Below, we are allowing Empty Approval Groups and assigning it to “Empty Approval Group”, that was created in Section 3.3.1.

SkipProjectApprover

IF

(for each case where) {

Journal Line is a Journal Line and

NOT(Journal Line.Entered Debit isn't null and

Journal Line.Entered Credit is null and

RL.string.substring(Journal Line.Account Combination,30,6) in "008815", "008902")

}

and

Lists is a Lists

THEN

List Builder Approval Group

Response Type Required

Approval Group Empty Approval Group

Allow empty groups True

Rule Name "SkipProjectApprover"

Dimension Id null

3.3.4 Org Approver Rule

For Debit Line, if Org is "00000" , "52101" , "53302" , "52101", assign to role ORG_<Org>_APPROVER

OrgAppr

Description

Effective Date Always

Priority Medium ☒ Active ☒ Advanced Mode ☐ Tree Mode

IF

(for each case where) {

Journal Line is a Journal Line and

Journal Line.Entered Debit isn't null and

RL.string.substring(Journal Line.Account Combination,4,5) in "00000", "52101", "53302", "52101"

}

THEN

List Builder	Resource
Response Type	Required
Users	null
Participants	Groups null
Application Role	"ORG_" + RL.string.substring(Journal Line.Account Combination,4,5) + "_APPROVER"
Rule Name	"OrgAppr"
Dimension Id	null

Below is the Role Definition and user assigned.

Edit Role ORG_00000_APPROVER: Users

☒ 1 Basic Information
 2 Function Security Policies
 3 Data Security Policies
 4 Role Hierarchy
 5 Segregation of Duties
 6 **Users**
 7 Summary

User Login	Display Name	Description
MLM77N		

3.3.5 Contra Org Rule

The Contra Rule specifies the action when the conditions are not met. Below, we are allowing empty approval groups and assigning it to “Empty Approval Group”, that was created in Section 3.3.1.

 ContraOrgAppr

IF

```

(for each case where) {
  Journal Line      is a Journal Line and
  NOT( Journal Line.Entered Debit      isn't null      and
    RL.string.substring(Journal Line.Account
      Combination,4,5)      in "00000", "52101", "53302"      )
}

```

THEN

List Builder	Approval Group
Response Type	Required
Approval Group	Empty Approval Group
Allow empty groups	True
Rule Name	"ContraOrgAppr"

3.3.6 Function Rule

For Journal line as Dr line and Function as "00424" , "00528" , "00132" , "00170" , "00520" , "00180" , "00505" , "00179" , "00590" , "00530", send it to FUNC_<Func Value>_APPROVER

FuncAppr

IF

(for each case where) {

Journal Line is a Journal Line and

RL.string.substring(Journal Line.Account Combination,10,5) in "00424", "00528", "00132", "00170", "00520", "00180", "00505", "00179", "00590", "00530" and

Journal Line.Entered Debit isn't null and

Journal Line.Entered Credit is null

}

THEN

List Builder Resource

Response Type Required

Users null

Participants Groups null

Application Role "FUNC_" + RL.string.substring(Journal Line.Account Combination,10,5) + "_APPROVER"

Rule Name "FuncAppr"

Dimension Id null

Below is the Role Definition and user assigned.

Edit Role FUNC_00590_APPROVER: Users

Basic Information 2 Function Security Policies 3 Data Security Policies 4 Role Hierarchy 5 Description of Duties 6 Users 7 Summary

Description	User Login	Display Name
FISCHLP		

3.3.7 Contra Function Rule

The Contra Rule specifies the action when the conditions are not met. Below, we are allowing empty approval groups and assigning it to "Empty Approval Group", that was created in Section 3.3.1

NOT(	RL.string.substring(Journal Line.Account Combination,10,5)	in "00520", "00180", "00505", "00179", "00590", "00530"	and
	Journal Line.Entered Debit	isn't null	and
	Journal Line.Entered Credit	is null	) or
	Journal Line.Entered Credit	isn't null	
}			

THEN

List Builder	Approval Group
Response Type	Required
Approval Group	Empty Approval Group
Allow empty groups	True
Rule Name	"NoFuncApprNeeded"
Dimension Id	null

3.3.8 Controller Rule

For Each Dr Journal Line, Approval should go to "Controller Approval Group". Currently Susan would receive all the Journal Line Approval notifications as last Approver.

Below is the Controller Approval Group definition.

Groups

+

-

×

Search

Controller

Controller Approval Group

Details

Name

Controller Approval Group

Members

↓

susan@hen

↓

ControllerAppr

IF

(for each case where) {

Journal Line is a Journal Line and

Journal Line.Entered Debit isn't null

}

THEN

List Builder Approval Group
Response Type Required
Approval Group Controller Approval Group
Allow empty groups True
Rule Name "ControllerAppr"
Dimension Id null

3.3.9 Contra Controller Rule

The Contra Rule specifies the action when the conditions are not met. Below, we are allowing Empty Approval Groups and assigning it to "Empty Approval Group", that was created in Section 3.3.1.

Contra

F

(for each case where) {

Journal Line is a Journal Line and

Journal Line.Entered Debit is null

}

THEN

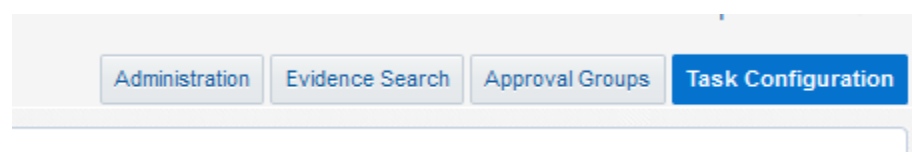
List Builder Approval Group
Response Type Required
Approval Group Empty Approval Group
Allow empty groups True
Rule Name "ControllerAppr"
Dimension Id null

3.3.10 Use Case: Adding New Function in BPM

Use Case depicts adding a new function in the existing BPM Approval. Assuming we have received a request to add new function "00530" for Approvals.

Steps Involved:

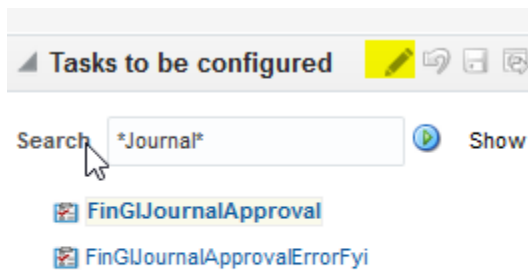
- 1) Navigate to BPM using BPM Worklist.
- 2) Go to Task Administration



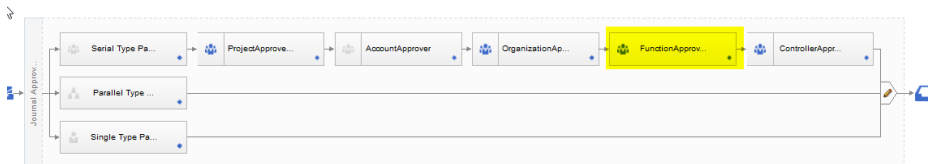
3) Search for Task related to Entity Approval



4) Edit the task (pencil icon)



5) Open the Function Approver Rule



6) Add New Function in the condition

FunctionApproverRuleSet

+

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✂

▼

is a Journal Line

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▼

tring(Journal Line.Account Combination

ntered Debit

ntered Credit

Test

Right Operand

"RL.string.substring(Journal Line.Account Combination,10,5) in"

Operand1	"00424"	🔍	,	+	-
Operand2	"00528"	🔍	,	+	-
Operand3	"00132"	🔍	,	+	-
Operand4	"00170"	🔍	,	+	-
Operand5	"00520"	🔍	,	+	-
Operand6	"00180"	🔍	,	+	-
Operand7	"00505"	🔍	,	+	-
Operand8	"00179"	🔍	,	+	-
Operand9	"00590"	🔍	,	+	-
Operand10	"00530"	🔍	,	+	-
Operand11		🔍	,	+	-

7) Make sure we add same condition in Contra Function

IF

(for each case where) {

Journal Line is a Journal Line and

☐ NOT(RL.string.substring(Journal Line.Account Combination in "00424", "00528", "00132", "00170", "00520", "00180", "00505", "00179", "00590", "00530"

☐ Journal Line.Entered Debit isn't null

☐ Journal Line.Entered Credit is null

☐ Journal Line.Entered Credit isn't null

}

8) Save and commit your changes

9) Create Custom Role FUNC_00530_APPROVER and add the Approver as user.

10) Test the rule by creating new Journal with Function and check for Approvals.

4. General Ledger

4.1 General Ledger Workflow Steps

Workflow	Approval Requirement
Journal Entry	1. The following sources require Approvals and the rest are auto-approved a. Manual b. Auto-Copy c. Recurring d. Spreadsheets
	2.Steps: i) Employee prepares a journal and attaches documentation if required. If the Journal source is one of the above, the approval is required; otherwise it is auto-approved. ii) The approval flow looks at the segment value of Location, Project, Function, Organization (in this order). on the debit line of the journal. <u>If there are multiple debit lines, the system will only look at the first segment values in the first line.</u> iii) If the segment value does not require approval from its owner, the journal will proceed to the next segment value

Workflow	Approval Requirement
	and so on. iv) At the end, the journal will go to the Controller for approval. v) Please refer to the spreadsheet in Sharepoint provided by Accounting for the names of the segment owners against all the segment values. (for the 4 segments mentioned above).

5. Procurement

5.1 Procurement Workflow Steps

Workflow	Module	Description / Steps
Supplier Registration	Procurement	1. Prospective Supplier selects Supplier Registration link via MCPS website. 2. Prospective Supplier Enters Data (Company Details, Contacts, Address, Business Classification, Products and Services, etc.) 3. Procurement Dept Reviews and Approves, Rejects or Request that the Supplier Resubmit 4. Payables Dept Reviews and Approves, Rejects or Request that the Supplier Resubmit The Requests to Resubmit action returns the request to the supplier, to complete missing data for resubmission. Suppliers cannot respond to rejected requests. Suppliers can register themselves through the supplier portal or can be invited by Purchasing for Bids. In both the cases, the supplier registration through the external portal will be routed for internal approvals.
Supplier Spend Authorized	Procurement	1. Execute Supplier Registration Steps.

Workflow	Module	Description / Steps
		2. Payables Dept promotes the Prospective Supplier to Spend Authorized. Payables upon completing their financial pre-requisite review process, will perform this approval action.
Supplier Profile Change	Procurement	1. Spend Authorized Supplier submits a request to change their Supplier Profile via the Supplier Portal 2.Accounts Payable Department (Supplier Administrator Role) will Review and Approves, Rejects or Requests More Information Primary: Grace Tsai, Andrew Tucci and Kim Houser Back-up: Susan Chen, Angela McIntosh-Davis and Rita Skolnik The Requests for More information returns the request to the supplier, who provides additional information via Comments and resubmits the request. Suppliers cannot respond to rejected requests.
Internal Supplier Registration	Procurement	MCPS has decided not to utilize Internal Supplier Registration
Solicitation	Advanced Procurement	MCPS has decided not to utilize the solicitation workflow
Contract	Advanced Procurement	MCPS has decided not to utilize the Contract Approval workflow

Workflow	Module	Description / Steps
Requisition	Procurement	1. Requisition is entered by MCPS requester 2. All NIGP codes are from FMS. The has been a consolidation; therefore, NIGP mapping is required. 3. Transaction is identified as an Internal Requisition or a Purchase Requisition 4. Internal Requisitions: a. Internal Requisitions only require the GL Account Owner Logic approver described in section 2 Above b. Reject Internal Requisitions with PATEO 5. Purchase Requisitions: a. Purchase Requisitions are identified as Project based or Non-project based b. If chart of account is PATEO, only the Project Manager defined in the P & G is required for approval (Sequence #1) c. Transactions utilizing GL account code combination 03-00000-00000-000-00-141040-000000-0000-00 (for warehouse / inventory use) are routed to Eugenia Dawson only for approval (Sequence #2)

Workflow	Module	Description / Steps
		d. GL string related transactions route to the GL Account Owner Logic described in section 2 Above (Sequence #3) e. Transactions utilizing natural account 212023 or 212083 are routed to Susan Chen for approval with an FYI notification to Kevin Soisson, regardless of dollar amount (Sequence #4) f. Transactions utilizing natural account 505070 are routed to Virginia Denning for approval with an FYI notification to Kevin Soisson, regardless of dollar amount (Sequence #5) g. FA transactions that are equal to or greater than 1K and less 5K route to Property Management (Sequence #6) h. FA transactions that are equal to or greater than 5K route to a Senior Accountant (Sequence #7) i. Requisition is routed via commodity code-based level 2 Manage Procurement Category Hierarchy mapping for category (NIGP Code) and Fixed Asset approvals. NIGP Code + Dollar Amount + Natural Accounts based approval. (Sequence #8)

Workflow	Module	Description / Steps
		j. Instruction Materials, technology, and school plant operations are routed to Subject Matter and Reference Material Approvers (Sequence #9) k. For Operating Grants / Title 1 utilize GL string and route per the Chart of Accounts details (Project Number), provided on the first line of the requisition. (Sequence #10) l. Transactions for Building Services are routed to the Area Supervisor and to the Director of Plant Operations for approval, based on NIGP Code (i.e., 485.00 and 485.65) (Sequence #11) m. Transactions associated with a punchout are routed to the Buyer, as the final approval (Sequence #12) The Request More Information and Reject actions, return the requisition to the Requisitioner for modifications. The Requisitioner can update and resubmit the transaction or cancel the transaction. Changes requested for a requisition associated with a purchase order, requires Buyer Approval.

Workflow	Module	Description / Steps
Purchase Order	Procurement	1. The dollar limits are based on PO approval volume, instead of Procurement Policy threshold. 2. A purchase order is entered / approved by a MCPS Buyer or MCPS Assistant Buyer 3. If the transaction is less than 15K, the MCPS Assistant Buyer and MCPS Buyer can approve transaction 4. If the transaction is equal to or greater than 15K, the MCPS Buyer can Approve, Reject or Request More Information. 5. If the Purchase Order is more than 15K, the MCPS Assistant Buyer, must reassigned to the appropriate MCPS Buyer for approval. The default Buyer is the Contract Administrator, if not specified. The Request More Information and Reject actions, returns the requisition to the Requester for modifications. The Requester can update and resubmit the transaction or cancel the transaction.

Workflow	Module	Description / Steps
Agreements	Procurement	1. An agreement is entered by a MCPS Buyer 2. If the transaction is less than 25K, the MCPS Assistant Buyer and MCPS Buyer can approve transaction 3. If the transaction is equal to or greater than 25K, the Buyer's job is identified as MCPS Assistant Buyer or MCPS Buyer. 4. If the job is MCPS Buyer, the Buyer can approve the transaction. 5. If the job is MCPS Assistant Buyer, the Assistant Buyer must be reassigned to the appropriate MCPS Buyer for approval. 6. The MCPS Buyer can Approve, Reject or Request More Information for transaction submitted by the MCPS Assistant Buyer.

6. Expenses

6.1 Expense Module Workflow Steps

Workflow	Description / Steps
MCPS Non-Local Travel	This process has multiple flows depending on selection criteria by Employee. 1) First Approver 2) Organization Approver Flow-1: First Approver only ===== Step 1: Employee creates Expense report. Step 2: Employee selects First Approver only By Default, Workflow Routing: First approver --> Organization Approver Flow-2: Organization Approver ===== Step 1: Employee creates Expense report. Step 2: Employee update the allocation organization segment, then it routes to the Organization Approver only Workflow Routing: Organization Approver (Final Approver)

Workflow	Description / Steps
	End result: Expense report Approved
MCPS Mileage Expenses	This process has multiple flows depending on selection criteria by Employee. First Approver Project Owner Program Owner Flow-1: First Approver + Project Approver + Function Approver ===== Step 1: Employee creates Expense report. Step 2: Employee selects First Approver + Project Approver + Function Approver Workflow Routing: First Approver → Project Approver -> Function Approver (Role Based) Flow-2 : First Approver + Project Approver ===== Step 1: Employee creates Expense report. Step 2: Employee selects First Approver + Project Approver Workflow Routing: First Approver → Project Approver -> (Role Based) Flow-3: First Approver ===== Step 1: Employee creates Expense report. Step 2: Employee selects First Approver only

Workflow	Description / Steps
	Workflow Routing: First approver -> Organization Owner Flow-4: Program Approver ===== Step 1: Employee creates Expense report. Step 2: Employee selects Program Approver only Workflow Routing: Program Approver (Final Approver) Flow-5: Project Approver ===== Step 1: Employee creates Expense report. Step 2: Employee selects Project Approver only Workflow Routing: Project Approver (Final Approver) End result: Expense report Approved Flow-6 : Default Approver ===== Step 1: Employee creates Expense report. Step 2: employee didn't update first approver or cost center Workflow Routing: Organization Approver (Final Approver) End result : Expense report Approved
Tuition Reimbursement	1) PDO team FTP File to MCPS folder

Workflow	Description / Steps
	2) Integration team to create Expense report for employees. 3) Expense report routes for approver to Template owner. 4) Exp Report Approved
DOT Expenses	1) DOT team sends file to AP team 2) AP team to FTP File to MCPS folder 3) Integration team to create Expense report for employees. 4) Expense report routes for approver to Template owner. 5) Exp Report Approved
DOM Expenses	1) DOM team FTP sends file to AP team 2) AP team to FTP File to MCPS folder 3) Integration team to create Expense report for employee 4) Expense report routes for approver to Template owner. 5) Exp Report Approved
DFNS Expenses	1) DFNS team FTP sends file to AP team 2) AP team to FTP File to MCPS folder 3) Integration team to create Expense report for employee 4) Expense report routes for approver to Template owner.

Workflow	Description / Steps
	5) Exp Report Approved
Project PATEO Expenses	All PATEO based expense reports are routed to Project Manager of the Project 1) Employee prepares expense report in system 2) Employee selects First Approver. 3) By Default, Expense reports routes to first approver and then to Project Manager of the Project. 4) Exp Report Approved
MCPS Grants Mileage Expenses	=== Same as MCPS Mileage Expenses===
MCPS Other Reimbursements	=== Same as MCPS Mileage Expenses===
MCPS DOC User	1) DOC team prepares expense report on behalf of employee in system and attaches documentation 2) Expense reports routes to Approver group – MCPS DOC Approver 3) Exp Report Approved

7. Accounts Payable

7.1 Accounts Payable Workflow Steps

Workflow	Description / Steps
Direct Invoice Routing	Flow-1: Requestor is not null Step 1: AP Team creates off-po invoices Step 2: Provides Requestor name in Invoice Header section Step 3: Validate Invoice Step 4: Initiate Workflow Workflow Routing: Workflow will be routed to Requestor Name provided. End result: AP Invoice Approved Flow-2: Requestor is same as Invoice creator Step 1: AP Team creates off-po invoices Step 2: Provides Requestor name (self) in Invoice Header section Step 3: Validate Invoice

Workflow	Description / Steps
	Step 4: Initiate Workflow Workflow Routing: Workflow will be Rejected End result: AP Invoice Rejected
PO Matched Invoice	No approval required, Invoice is Auto approved
OFF PO Invoice	For Auto-Approval process, AP team should not populate Requestor name in Invoice Header section
Invoice created through Supplier Portal	Step 1: Supplier creates off-po invoices through Supplier Portal Step 2: Provides Requestor name in Invoice Header section Step 3: Submits Invoice Request Step 4: Invoice Request Auto-approved Step 5: AP team Validate Invoice Step 6: Initiate Workflow Workflow Routing: Workflow will be routed to Requestor Name provided. End result: AP Invoice Approved

8. Projects & Grants

8.1 Projects & Grants Workflow Steps

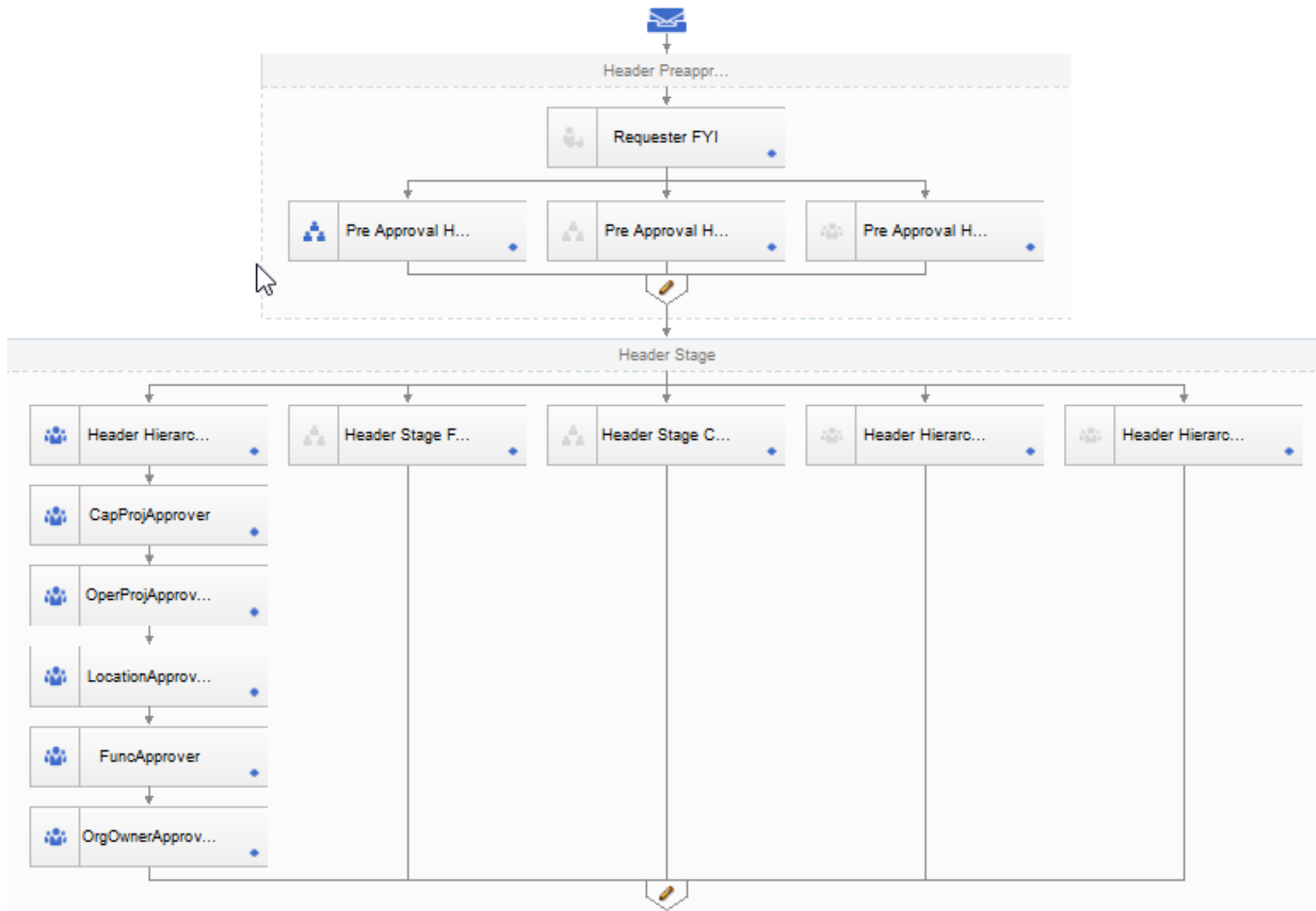
Workflow	Description / Steps
Create Project	1) Multiple individuals may create projects; however, a Project Approver Role (Theresa or Rachel) must approve creation and completion.
Create Award Budget	1) Routes to Principal Investigator of the Award (Terry Miller)
Contract Approver	1) Auto creation and Auto Approve of Contract while Award definition and submission for Approval. 2) Approval of Contract activates the Award. 3) Auto Approve for Amendment of Contract.

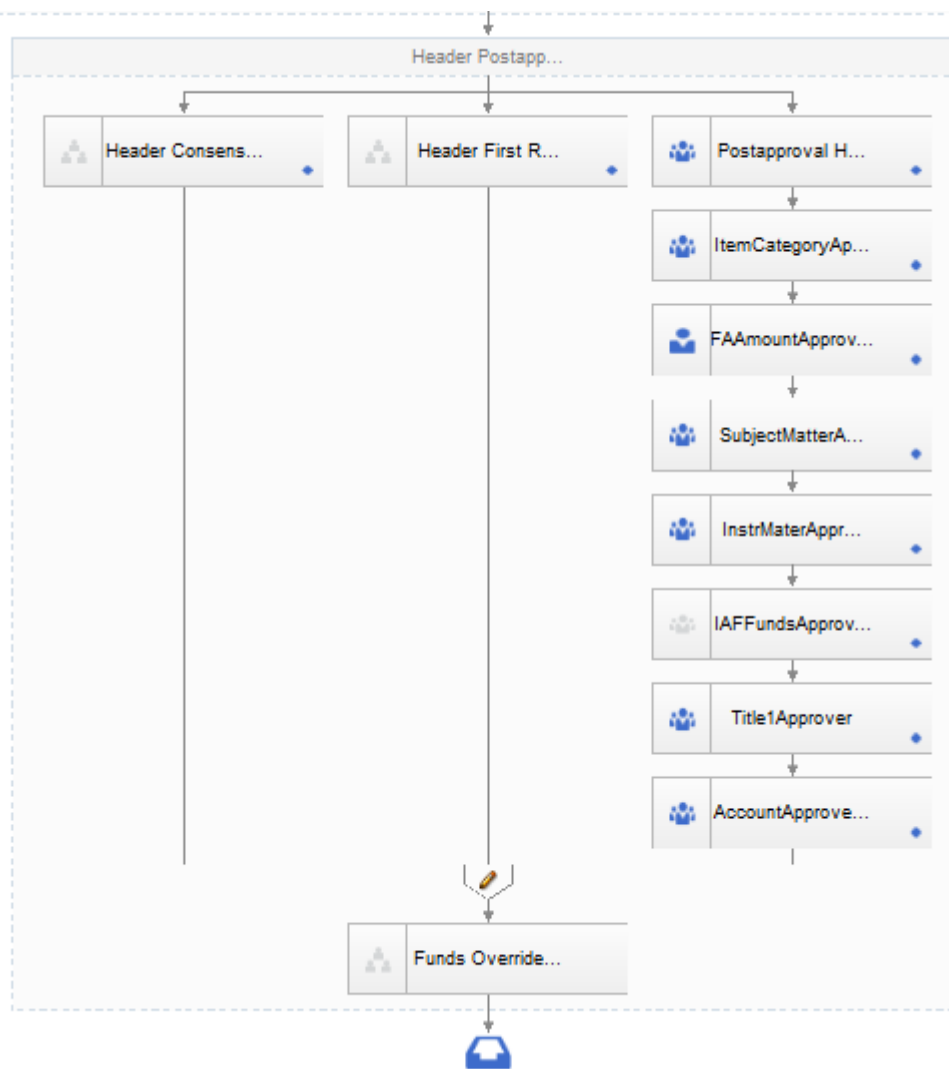
9. Technical Overview

9.1 Requisition Approval Workflow (ReqApproval)

Below is the flow chart (Vertical) for ReqApproval.

Assignees





9.1.1 PreApprovalHeaderConsensusRules

9.1.1.1 Rule: POETARequired

This Rule states that if the Charge Account Starts with “04” and doesn’t have Project associated, Reject the Requisition.

POETARequired

IF

Each {

ReqDistributionDimension is a ReqDistributionDimension and

ReqDistributionDimension.chargeAccount starts with "04" and

ReqDistributionDimension.projectId is null

}

and

Task is a Task

THEN

List Builder Supervisory
Response Type Required
Number of levels 1

Starting Participant `HierarchyBuilder.getPrincipal(Task.creator,-1,"")`

Top Participant `HierarchyBuilder.getManager("supervisory",Task.creator,-1,"")`
Auto Action Enabled True

Auto Action "REJECT"
Rule Name "POETARrequired"

Dimension Id null

9.1.1.2 Rule: NoPOETARrequired

This is skip rule for Rule **POETARrequired**

```

IF
.....

Each {
  ReqDistributionDimension is a ReqDistributionDimension and
  NOT( ReqDistributionDimension.chargeAccount starts with "04" ) or
  (   ReqDistributionDimension.projectId isn't null and
    ReqDistributionDimension.chargeAccount starts with "04" )
}

and
Lists is a Lists
.....

THEN

call ▼ IgnoreParticipant ( "PreApprovalHeaderConsensusRules", Lists )

```

9.1.2 HeaderHierarchyRules

9.1.2.1 Rule: MoveontoNextRule

This is blank rule created to test the workflow.

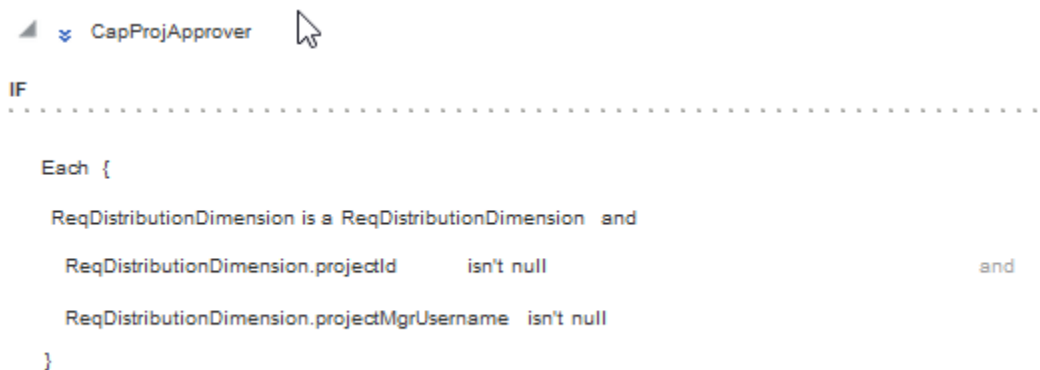


9.1.3 CapProjApprover

9.1.3.1 Rule: CapProjApprover

This rule states below:

- 1) If Project Id is not null and Project Manager is not null send Approval to Project Manager.



- 2)

THEN

List Builder Resource
Response Type Required
Users ReqDistributionDimension.projectMgrUsername
Participants Groups null
Application Role null
Rule Name "CapProjApprover"
Dimension Id null

9.1.3.2 Rule: NoCapPrjMgrReject

- 1) If Project Id is not null and Project Manager is null , Reject the Requisition



.....

THEN

List Builder	Supervisory
Response Type	Required
Number of levels	1
Starting Participant	HierarchyBuilder.getPrincipal(Task.creator,-1,"")
Top Participant	HierarchyBuilder.getManager("supervisory",Task.creator,-1,"")
Auto Action Enabled	True
Auto Action	"REJECT"
Rule Name	"POETARRequired"
Dimension Id	null

9.1.3.3 Rule: SkipCapProjApprover:

If Project is not present for REQ, ignore the rule.

SkipCapProjApprover

IF

None {

ReqDistributionDimension is a ReqDistributionDimension and

ReqDistributionDimension.projectId isn't null

}

and

Lists is a Lists

THEN

call IgnoreParticipant ("CapProjApprover", Lists)

9.1.4 OperProjApprover

9.1.4.1 Rule: OperProjApprover

This rule is based on Requisition Distribution Project. Its currently not being used and can be knocked down if no requirements come.

▲ ▼ OperProjApprover

IF

Each {

ReqDistributionDimension is a ReqDistributionDimension and

RL.string.substring(ReqDistributionDimension.chargeAccount,30,6) isn't "000000"

}

and

Lists is a Lists

THEN

call ▼ IgnoreParticipant ("OperProjApprover", Lists)

9.1.4.2 Rule: SkipOperProjApprover



9.1.5 LocationApprover

9.1.5.1 Rule: LocationApprover

This rule is based on Requisition Distribution Location. Its currently not being used and can be knocked down if no requirements come.

Rule: LocationApprover

LocationApprover

IF

Each {

ReqDistributionDimension is a ReqDistributionDimension and

RL.string.substring(ReqDistributionDimension.chargeAccount,16,3) isn't null

}

and

Lists is a Lists

THEN

call IgnoreParticipant ("LocationApprover", Lists)

9.1.5.2 Rule: SkipLocationApprover



9.1.6 FuncApprover

9.1.6.1 Rule: FuncApprover

This Rule States:

- 1) If Function is in "00424" and "00528" and Project is null, Send Approval to FUNC_<FUNC VALUE>_APPROVER.

▼ FuncApprovers

IF

Each {

ReqDistributionDimension is a ReqDistributionDimension and

RL.string.substring(ReqDistributionDimension.chargeAccount,10,5) in "00424" , "00528" and

ReqDistributionDimension.projectId is null

}

THEN

List Builder Resource

Response Required

Type

Users null

Participants Groups null

Application Role "FUNC_"+RL.string.substring(ReqDistributionDimension.chargeAccount,10,5)+"_APPROVER"

Rule Name "FuncApprovers"

Dimension Id null

9.1.6.2 Rule: SkipFuncApprovers

This rule is Skip condition for Rule FuncApprover which specifies ignore the rule.



9.1.7 OrgOwnerApprover

9.1.7.1 Rule: OrgOwnerApprover

This rule is based on below condition:

If Project is null and Distribution combination is not equal to 03-00000-00000-000-00-141040-000000-0000-00 , Approval will go to Cost Center Manager defined at Department level.

OrgOwnerApprover

IF

Each {

ReqDistributionDimension is a ReqDistributionDimension and

ReqDistributionDimension.projectId is null and

ReqDistributionDimension.codeCombinationId isn't 11726

}

THEN

List Builder Resource

Response Type Required

Users ReqDistributionDimension.costCenterManagerUsername

Participants Groups null

Application Role null

Rule Name "OrgOwnerApprover"

Dimension Id null

9.1.7.2 Rule: InvOrgOwnerApprover

If Project is null and Distribution combination is equal to 03-00000-00000-000-00-141040-000000-0000-00 , Approval will go to Static Approval Group "INVENTORY".


InvOrgOwnerApprover

IF

Each {

ReqDistributionDimension is a ReqDistributionDimension and

(ReqDistributionDimension.projectId is null and

ReqDistributionDimension.codeCombinationId is 11728)

}

and

Lists is a Lists

THEN

List Builder Approval Group

Response Type Required

Approval Group INVENTORY

Allow empty groups False

Rule Name "InvOrgOwnerApprover"

Dimension Id null

9.1.7.3 Rule: SkipOrgOwnerApprover

This is skip condition for this Rule and would ignore the rule.

SkipOrgOwnerApprover

IF



Each {

ReqDistributionDimension is a ReqDistributionDimension and

(ReqDistributionDimension.projectId isn't null)

}

and

Lists is a Lists

THEN

call IgnoreParticipant ("OrgOwnerApprover", Lists)

9.1.8 PostApprovalHeaderHierarchyRules

9.1.8.1 Rule: PostApprovalBegin

This rule is just to begin Post Approval Workflow. We are ignoring it.

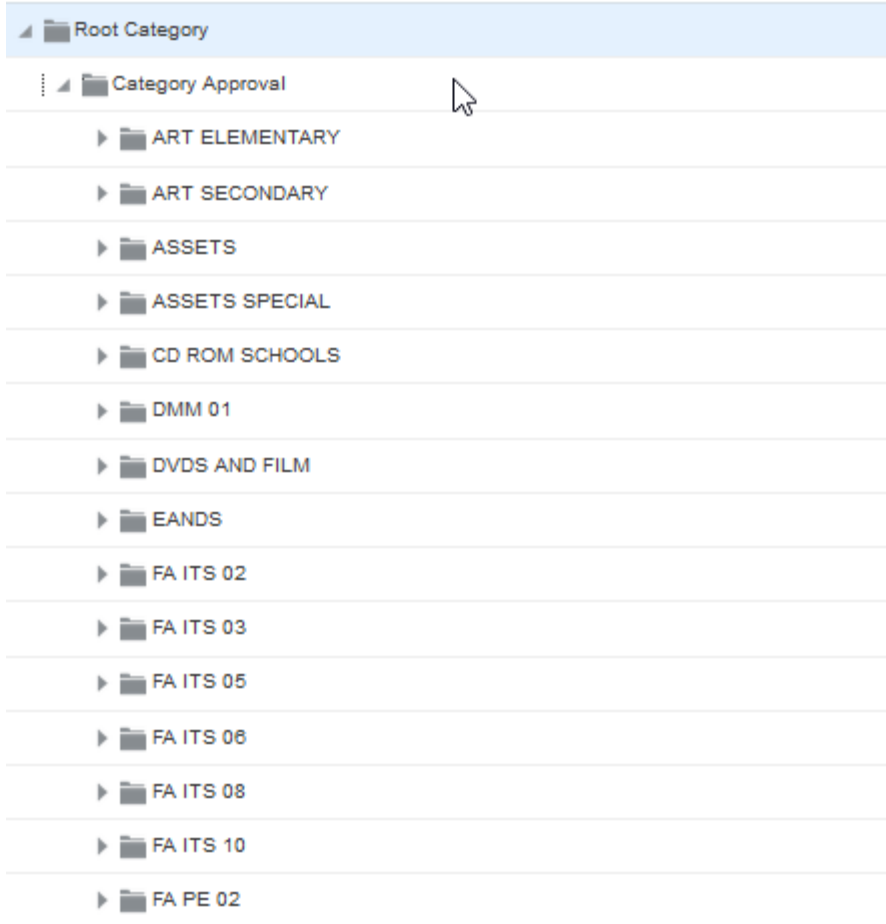


9.1.9 ItemCategoryApprovers

All the rules follow below common conditions:

1. Req shouldn't be associated with Project.
2. Req is not tied to Inventory Code Combination "03-00000-00000-000-00-141040-000000-0000-00"

All the Categories are tied to parent folder (Level-2) based on Approval requirement in "[Manage Procurement Category Hierarchy](#)"



For all Categories which doesn't require approval are put in Default folder.

 Default
 MCPS Categories
 005.28 Abrasives, Sandblasting (Other than Metal)
 005.56 Abrasives, Tumbling (Wheel)
 005.63 Grinding and Polishing Compounds: Carborundum, Diamond, etc. (For Valv...
 005.70 Pumice Stone
 005.75 Recycled Abrasives Products and Supplies
 005.84 Steel Wool, Aluminum Wool, Copper Wool, and Lead Wool

9.1.9.1 Rule: DMM 01

This rule is based on below condition:

10. Open and Closed Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date
1	Approver Names not getting displayed on the transaction inquiry.	A custom report built	AST		
2	Technical Design to be updated after functional design approved		Deepak		